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Cécily Defoort

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CÉCILY DEFOORT*

Long-term Trends in International Migration: An Analysis of the Six Main Receiving Countries

In a global context of growing international mobility and rising education levels, emigration by highly skilled individuals is, not surprisingly, on the increase. To observe the brain drain, an analysis by individual sending countries is more instructive than a general overview. In this article, Cécily DEFOORT uses an OECD database on immigrant stocks by skill level in the six main OECD receiving countries to reconstitute country-specific trends in emigration rates among highly skilled workers over the period 1975–2000. Small countries, like those of the Caribbean and Oceania, are especially affected. Their lack of university infrastructure makes them particularly vulnerable to the permanent loss of students who emigrate for study purposes and do not return home. Among larger countries, some traditional emigration flows continue throughout the study period, affecting countries like Kenya and Uganda, for example, though also the United Kingdom. The countries identified as badly affected in 2000, but not in 1975, include large countries of Africa and South Asia. On the other hand, although vast nations like China and India export large numbers of skilled migrants, their seemingly inexhaustible reserves of human capital mean that the local impact of these losses is slight.

International migration has increased considerably since the 1960s. According to the UN, the number of people living outside their country of birth rose from 75 million in 1960 to 175 million in 2000 (United Nations, 2002). Because of their consequences for sending and receiving countries alike, the impact of these flows has always attracted attention from political decision-makers. A key question, with far-reaching implications for development in the countries of origin, is that of migrant skill levels.

The literature on the impact of brain drain on sending countries began to expand during the 1960s. A first wave of research viewed migration by skilled workers in an optimistic light as being beneficial in the long run. Assuming

* EQUIPPE, Universities of Lille (France) and IRES, Université catholique de Louvain (Belgium).
Translated by Godfrey Rogers.

perfectly competitive markets, it was argued that phenomena such as remittance inflows, return migration and the development of networks could offset the negative short-term effects associated with loss of the most skilled individuals (Grubel and Scott, 1966; Johnson, 1967).⁽¹⁾ However, in a second wave of research that developed over the next twenty years (1970s and 1980s), the authors emphasized the adverse consequences for the countries of origin, notably the net loss of the educational investment in their native populations (Bhagwati and Hamada, 1974; Usher, 1977; Blomqvist, 1986; Ul Haque and Kim, 1995). A particular feature of this period was the proposal to create a “brain drain tax”, dubbed the Bhagwati tax (Bhagwati, 1972), to be levied on emigrants living in developed countries with the aim of compensating the poorest countries for their human capital losses. Emigration was thus seen as threat to economic growth in sending countries, with the potential to curb their future development prospects.

A further change of direction in the research literature occurred in the 1990s. Economists once again emphasized the potentially positive effects of brain drains, hitherto judged marginal. To the “classic” externalities of skilled migration were added the concepts of “migration perspectives” and “migrant selection” in developed countries. Economists demonstrated that emigration by highly skilled workers could, under certain conditions, have a beneficial impact for the countries of origin, in particular when individuals decide to acquire a skill with a view to emigrating at a later date but, for various reasons, are then unable to leave their country of origin. This turnaround is abundantly documented in the literature on skilled migration (Stark et al., 1997; Mountford, 1997; Beine et al., 2001, 2008).

Empirical approaches, for their part, have been hampered by a lack of standardized and continuous data on international migration. Difficulties are linked notably to the problem of establishing an international definition of migrant status and to the emergence of migrant flows from new countries following the collapse of the Soviet regime. Important progress was made when Carrington and Detragiache (1998, 1999), two researchers at the IMF, estimated rates of emigration to the OECD countries from 61 developing countries for 1990, distributed across three skill levels (using Barro and Lee’s (2000) data on capital human and those of the OECD for migration). This database marks the starting point for the empirical study of brain drains. Adams (2003) subsequently introduced the temporal dimension by supplying emigration data for 24 sending countries in 2000. Using census data for the OECD countries, Dumont and Lemaître (2005) calculated emigration rates by skill level for around 100 sending countries in 2000. In the same vein, Docquier and Marfouk (2005) produced new estimates of numbers and emigration rates by level of skill for 195 sending countries in 2000 and 175 countries in 1990. These data can be used to test empirically the theories of international migration (link

(1) For more details, see the literature review by Docquier and Rapoport (2005).

between migration and direct investment abroad, migration and network effects, impact of brain drain on human capital, etc.). Altogether, these databases, like the UN estimates, show that emigration rates have risen continuously over the last thirty years, at speeds that vary between world regions, and the phenomenon is attracting growing interest among researchers.⁽²⁾

The main purpose of this article is to present a new database on international migration constructed using a method similar to that used by Carrington and Detragiache (1998, 1999) and Docquier and Marfouk (2005). We assembled data on resident migrants in the six main OECD receiving countries (United States, Canada, Australia, United Kingdom, Germany and France) and on the structure by skill level of emigrant and resident populations for all countries of the world between 1975 and 2000 (five-yearly observations), covering a longer period than was the case in previous studies. Since current migrant stocks depend on past and present migration decisions, a database organized on chronological lines is particularly useful for empirical research.

After describing the method used to construct the chronological database, we present the long-term trends in human capital and in migration of skills by world region. We then analyse the changes in migration rates of highly skilled individuals by country between 1975 and 2000. Last, we focus our attention on the countries where the impact from migration of the most skilled workers is greatest relative to population size.

I. Data and method

In this section we describe the data and the method used to estimate the stocks and emigration rates from all countries of the world since 1975. Stock variables are more appropriate for analysing the endogeneity and dynamics of migratory movements (the equilibrium values are always given as stocks, not flows). From a statistical viewpoint, migration flows are less reliable than stocks, since return migration (outflows) is impossible to quantify.

The method employed is in three stages. We begin by estimating immigrant numbers in the main OECD receiving countries by skill level and country of origin. In a second stage, this information is used to reconstruct stocks of emigrants by country of origin. Lastly, with these figures for emigrant numbers and the resident population size in the sending countries, we derive estimates of emigration rates for each sending country. These three stages involve a set of assumptions that must be described in detail.

(2) No comparative data on international migration by skill level are available for South-North migration. We discuss the consequences later in this article.

1. Numbers of international migrants

To reconstruct emigrant numbers by skill level we first use information about the number of immigrants aged over 25 in the main receiving countries. As it is practically impossible to quantify emigration from every country in the world, we use the figures for immigrant numbers from the population censuses of the six main OECD receiving countries: Australia, Canada, United States, United Kingdom, France and Germany (OECD-6 hereafter). These are the six countries for which data are available for the entire period from 1975 (including for Germany with its current borders). Our principal focus is on South-North migration.

Immigrant numbers are normally available by country of birth, except for Germany where the breakdown for 1980, 1990 and 2000 is by nationality. The information for Germany comes from the *Mikrozensus* of the Statistisches Bundesamt (Federal Institute of Statistics). Using citizenship rather than country of origin may lead to overestimating migrant skill levels: in principle, a German-born adult of foreign nationality (a Turk, for example) will on average have higher skills than a Turkish-born individual who moved to Germany as an adult. Furthermore, the number of individuals applying for German nationality rises with skill level.⁽³⁾ This latter factor may lead to underestimating the stock of skilled migrants in the receiving country. Although it is hard to quantify the effect of each of these factors, the influence of the former appears to outweigh that of the latter. We can therefore assume an overestimation of the number of skilled immigrants in Germany that cannot be corrected using the available data.

Migrants living in the OECD-6 countries represent approximately 77% of aggregate immigrant populations in the OECD countries in 2000, so our estimates can be taken as representative of total South-North immigration. The potential bias introduced by this restriction will be discussed in the next section.

The years for which information is not available (between censuses) are handled by interpolation (1995 and 1985) or retropolation (1975). To perform the interpolations we use the OECD complementary data (SOPEMI, 2005). These data are generally less detailed than the censuses, but they give total immigrant numbers for 1980-1990 and for 1990-2000, and in some instances a distribution by region. Knowing the total immigrant stock for the intermediate years (1985 and 1995), we calculate the worldwide numbers according to skill level in each destination country by multiplying the total by the proportion of highly (and medium) skilled persons among the immigrants (hm_i). This proportion is defined in relation to the proportions for the previous and

(3) Acquisition of Germany nationality is subject to seven conditions: eight years of legal residence in Germany, possession of a residence permit, acceptance of the Basic Law, absence of anti-constitutional activities, no dependence on welfare or unemployment benefit, absence of criminal record excepting minor offences, adequate knowledge of the German language.

subsequent years: $(t - 1)$ and $(t + 1)$. If we denote as hm_t the proportion of highly skilled persons among the immigrants at date t (this proportion is known for 1980, 1990 and 2000), and as hr_t the proportion of highly skilled persons among the resident population at the same date (proportion known for every period), the proportions of highly skilled persons among migrants at the intermediate dates 1985 and 1995 (at t) are given by:

$$hm_t = \frac{\left(\frac{hm}{hr}\right)_{t-1} + \left(\frac{hm}{hr}\right)_{t+1}}{2} hr_t$$

Thus we obtain the total number of highly (and medium) skilled immigrants. This stock figure is then distributed by country of origin. The number of skilled emigrants from country j in host country i at date t is given by the number of skilled immigrants from all countries in country i multiplied by the proportion of emigrants from country i relative to total emigration. This proportion depends on the proportions observed at $(t - 1)$ and $(t + 1)$. To ensure that the numbers by country of origin sum to exactly the world total we assign weights to the proportions observed at $(t - 1)$ and $(t + 1)$.⁽⁴⁾ The same method is employed for 1975, though calculation is based on the later year (1980) only.

Three levels of qualification are distinguished: h for highly skilled, m for medium skilled, and l for low skilled. A brain drain corresponds to emigration by the highest skilled workers. Workers designated as low skilled have a primary-level education (0-8 years of education); medium skilled workers have a secondary qualification (9-12 years of education); and the highly skilled have a tertiary qualification (13 years or more of education). These data are drawn from the population censuses or registers of the host countries. The categories are constructed using information specific to the countries and are compatible with the human capital indicators available for all countries of origin. The criterion for ranking by skill level is the same for all countries.

$Z_{s,t}^{i,j}$ denotes the stock of immigrants in country i ($i = 6$) originating from country j ($j = 172$) in year t ($t = 1975, \dots, 2000$), of skill level s ($s = l, m, h$).

Table 1 summarizes the data sources used to estimate immigrant numbers in the main OECD receiving countries since 1975.

If we adopt the standpoint of the countries of origin, we need to aggregate the information from the six OECD countries under consideration. This gives us the emigrant stocks from all countries in the world: $M_{s,t}^{i,j}$.

(4) The procedure does not assign 0.5 to $(t - 1)$ and 0.5 to $(t + 1)$, but a weight that gives a total stock of qualified migrants exactly equal to that calculated previously (0.53 for $(t - 1)$ and 0.47 for $(t + 1)$, for example).

Table 1. Database sources

	Australia	Canada	United States	France	United Kingdom	Germany
1975	Census Australian Bureau of Statistics	–	–	Recensement 1975 (INSEE)	–	–
1980	Australian Bureau of Statistics	Census	Census (IPUMS)	Recensement 1982 (INSEE)	Census	Statistisches Bundesamt
1985	Census Australian Bureau of Statistics	OECD 2005 (SOPEMI report)	–	–	–	Mikrozensus
1990	Census ^(a)	Census ^(a)	Census (IPUMS) ^(a)	Recensement 1990 ^(a)	Census ^(a)	Mikrozensus ^(a)
1995	Census Australian Bureau of Statistics	Statistics Canada	US Census Bureau	–	–	Mikrozensus
2000	Census ^(a)	Census ^(a)	Census ^(a)	Recensement 1999 ^(a)	Census ^(a)	Mikrozensus ^(a)
^(a) For 1990 and 2000 we used the data given in Docquier and Marfouk (2004). – When data are missing we performed interpolations or retropolations.						

The stock of emigrants with skill level s , originating from country j , in year t , is obtained as follows: $M_{s,t}^j = \sum_{i=1}^6 Z_{s,t}^j$

Our method has several limitations:

- First, owing to non-availability of data, not every OECD receiving country is included in our database. Since we focus essentially on South-North migration, this introduces a slight bias into our estimates.⁽⁵⁾
- Second, our database does not include illegal immigration. According to the US Immigration and Naturalization Service, the illegally resident population in the United States numbered 3.5 million in January 1990 and 7 million in January 2000. While the principal country of origin can be established for most of these migrants (nearly 70% came from Mexico in 2000), their skill level remains unknown. Data for the other OECD countries are non-existent or extremely unreliable,⁽⁶⁾ and hence have not been included in our database. Assuming that the majority of illegal migrants are low skilled, it is likely that our database overestimates the proportion of highly skilled migrants.

(5) The next section contains a detailed explanation and discussion of this bias in the estimates.

(6) For example, the report by the French Senate, entitled *Immigration clandestine : une réalité inacceptable, une réponse ferme, juste et humaine* ["Illegal Immigration: an unacceptable reality, a firm, fair and humane response"], highlighted a wide discrepancy between the French Interior Ministry's figures and those of INED: in 1998, the former estimated that 80,000-100,000 additional illegal immigrants were entering France each year, whereas the latter estimated the annual net flow of illegal immigrants at around 13,000.

• Third, we have no information on age at entry to the host country. Consequently we cannot distinguish immigrants who were educated in their country of origin from those educated in their destination country. This is a major drawback, since it is difficult to see how an individual born in Côte d'Ivoire who comes to France to study can represent an example of brain drain. Restricting the database to individuals aged 25 and over (see below) does, however, exclude people who migrated solely for study purposes (because there was no higher education in their home country, for example, as is the case in many of the small islands badly affected by emigration). Once again, the data supplying age at entry to the host country are not available for the entire period under consideration. The only such available data come from Beine et al. (2007). They use a gravity model to estimate the age at entry of migrants, distinguishing migration after age 12, after age 18 and after age 22 for immigrants in the OECD countries for which this information is not available (23%). Age at entry is the basis for determining whether education was received in the sending or the receiving country. These estimates are available from 1990 and show a close correlation between the data by birth country and the data by country where skills were acquired: the countries hardest hit by brain drain identified from sending country data are on the whole the same as those identified from receiving country data. This result suggests that the bias will also be small in our database, which is built using the same methodology as that of Docquier and Marfouk (2005) and does not include the country where skills were acquired.

• Fourth, we restrict attention to the population of working age (age 25 or over). There are two reasons for this choice. First, when calculating emigration rates we use the stocks of resident population, which, as we shall see later, are only available for the over-25s. Another reason is that working on stocks of migrants aged 25 or over, rather than on total migrant numbers, removes from the database those individuals who migrate solely to study and who go straight back to their country of origin after completing their education. This decision limits the bias introduced by migrants being considered by their country of birth rather than by the country where they were educated (see above). Confining attention to the population aged 25 and over is one reason why our estimates are lower than those of the UN.⁽⁷⁾

• Fifth, a powerful critique can be made regarding comparability of skill levels worldwide. A diploma in medicine or financial management may have a particular value in the US, but it is uncertain that the same diploma can be treated as exactly equal when awarded in a country whose educational system is relatively under-developed. Coulombe et al. (2004) addressed this problem. Data from the 1994 International Adult Literacy Survey (IALS) was used to study the link between human capital and economic growth in the OECD

(7) In 2000, the UN estimates at 81 million the number of immigrants resident in the OECD countries, all age groups combined (UN, 2002); Docquier and Marfouk (2005) put the number at 57 million by counting only individuals aged 25 or over. Thus the under-25s correspond to approximately 30% of the total immigrant population in the OECD countries.

Table 2. Descriptive statistics: migrants aged 25 or over resident in the six main OECD receiving countries and percentage distribution by skill level, 1975-2000

	Australia	Canada	United States	France	United Kingdom	Germany
1975						
Number	2,012,942	2,764,850	7,805,413	2,989,461	2,183,263	2,174,924
Low skilled	48.1	50.2	36.5	92.9	78.7	85.1
Medium skilled	20.7	9.4	38.2	3.1	10.0	6.6
Highly skilled	31.2	40.4	25.3	4.0	11.3	8.2
1980						
Number	2,205,469	2,860,690	9,522,320	3,162,661	2,362,192	2,408,330
Low skilled	41.4	46.8	34.9	91.2	72.2	82.7
Medium skilled	25.8	8.5	35.5	3.4	14.9	7.5
Highly skilled	32.7	44.7	29.6	5.4	12.9	9.7
1985						
Number	2,589,429	2,991,971	11,788,017	3,269,569	2,529,953	2,364,218
Low skilled	37.8	40.4	30.7	88.4	70.2	78.4
Medium skilled	26.9	10.3	34.2	4.6	13.3	9.0
Highly skilled	35.4	49.3	35.1	7.0	16.5	12.6
1990						
Number	2,866,450	3,481,720	14,227,826	3,387,511	2,697,714	2,652,651
Low skilled	34.9	37.1	25.7	85.5	68.1	72.7
Medium skilled	29.3	12.0	34.0	5.9	11.3	10.9
Highly skilled	35.8	50.9	40.3	8.6	20.6	16.4
1995						
Number	3,062,474	4,011,267	17,562,694	3,636,495	3,115,599	4,363,464
Low skilled	35.0	29.9	36.4	79.4	51.7	71.3
Medium skilled	28.7	11.7	23.7	8.1	20.0	10.2
Highly skilled	36.3	58.4	39.8	12.5	28.4	18.5
2000						
Number	3,416,147	4,600,970	24,190,881	3,748,332	3,533,725	4,688,837
Low skilled	24.2	29.6	23.1	74.6	36.5	66.4
Medium skilled	33.2	11.6	34.4	9.0	28.8	12.3
Highly skilled	42.6	58.8	42.5	16.4	34.7	21.2
Source: Author's calculations based on census data.						

countries, with measures of skills based on “functional literacy”⁽⁸⁾ rather than on diploma levels or years of education. Their results are more convincing

(8) Statistics Canada observes: “Traditionally, literacy referred to the ability to read, understand and use information. However, the meaning of the term has broadened to encompass a range of knowledge, expertise and skills that relate to reading, mathematics, science and more besides”. The IALS survey is designed to evaluate the abilities of people aged 16-65 in three areas of “literacy”: comprehension of prose, quantitative and documentary texts (in 14 countries).

than those from earlier studies using the “traditional” human capital indicators. Unfortunately, these data are not available for all countries of the world for the period 1975-2000. Consequently we use Barro and Lee’s (2000) indicators for the non-OECD countries and those of De la Fuente and Domenech (2002) for the OECD countries.

Table 2 presents selected descriptive statistics on immigrant numbers in the six main OECD receiving countries for the period 1975-2000.

The largest increase in the immigrant population over the period occurred in the US, where it grew three-fold from nearly 8 million in 1975 to 24 million in 2000. There was also a large increase in Germany, where migrant numbers more than doubled between these dates. The stock of migrants increased by a factor of 1.6 on average in the other countries, and by a factor of 1.3 in France, the country where their numbers have increased least since 1975. We also observe a general increase in migrant skill levels in all six host countries. The proportion of skilled persons among immigrants is highest in Canada, where it is close to 60% (and increased by nearly 20 percentage points over 25 years). The proportion is 42% in 2000 in Australia and the United States. France and Germany are the countries where it is lowest: 16% and 21% respectively.

In the next section we compare our results with the estimates by Docquier and Marfouk (2005) to assess the reliability of our estimates.

2. Reliability of migrant stocks measured for the six main receiving countries

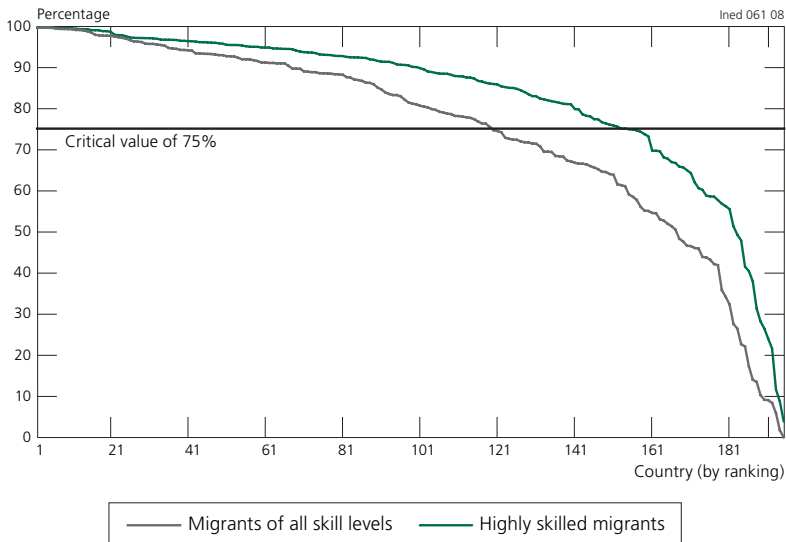
As noted at several points, figures for emigrant stocks by country of origin are derived from data on immigrant stocks in the six main OECD receiving countries. They do not represent the totality of migrant destination countries, however, since much international migration occurs between countries of the South. Migration to non-OECD countries generally involves unskilled migration, excepting for a few countries including South Africa (1.3 million immigrants in 2000), six Gulf states (9.6 million immigrants in Saudi Arabia, United Arab Emirates, Kuwait, Bahrain, Oman, and Qatar), plus a few countries in East Asia (4 million immigrants in Hong Kong and Singapore). However, according to Docquier and Marfouk (2005), skilled emigration to the OECD countries as a whole accounts for 90% of skilled migrants in the world. Since the six main receiving countries in our database attract 86% of skill migration, we therefore capture over 77% of global skilled emigration. Our database underestimates emigrant numbers for some twenty countries of origin (Egypt, Sudan, Jordan, Yemen, Pakistan, Namibia, Zimbabwe, Democratic Republic of Congo, Suriname, Mozambique, Angola, etc.), though with data from national statistical offices it is not possible to improve the base’s cross-sectional dimension while at the same time maintaining its longitudinal coverage.

Docquier and Marfouk (2005) conducted the same kind of study on immigrant numbers in the OECD countries as a whole (for 1990 and 2000). Depending on the country, therefore, they obtain stock figures considerably higher than ours. The reliability of our results can be assessed by comparing the migrant stocks obtained by Docquier and Marfouk in 2000 with the stock figures we obtain for the six main OECD countries at the same date. The comparison is expressed in percentage terms. For example, 84% of migrants originating from Nigeria present in an OECD country in 2000 live in an OECD-6 country. In other words, our database covers 84% (and not 100%) of Nigerian migration to countries in the north. The proportion is 94% if only skilled migrants are considered.

Figure 1 represents emigrants aged 25 and over to the OECD-6 countries as a proportion of total emigrants to OECD countries and likewise for highly skilled individuals, for each country of origin. The countries are ranked in decreasing order.

Of 195 countries, 156 can be considered “reliable” for measuring skilled emigration in that their skilled migrants resident in the OECD-6 countries represent a minimum of 75% of skilled migrants in the OECD countries as a whole in 2000 (unbroken line on figure). The countries “reliable” at the 80% and 70% levels number 141 and 162, respectively. Notable among the

Figure 1. Proportion of adult migrants present in OECD-6 countries with respect to all migrants in OECD in 2000



Source: Author's calculations based on census data and Docquier and Marfouk (2004).

countries considered to be “non-reliable” are Suriname (96% of skilled emigrants living in The Netherlands), Equatorial Guinea (emigrants concentrated in Spain), or Angola (emigration centred on Portugal). This group consists mainly of European countries (EU-25 and other countries of Eastern Europe).

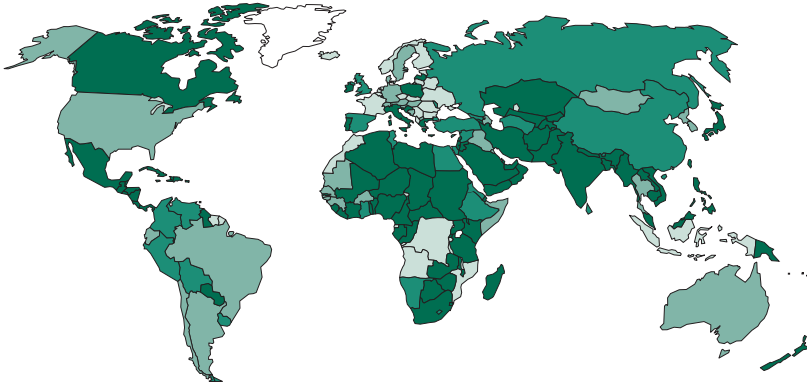
When emigrants of all skill levels are taken into account, Figure 1 (grey line) shows that the degree of reliability is lower. The bias inherent to our database therefore affects mainly unskilled migration. This is not an unexpected result considering that three of our six main receiving countries (Australia, Canada, the United States) have operated openly selective migrant admission policies for at least twenty years. Of 195 countries, 119 can be considered “reliable” for measurement of overall emigration to OECD countries, with their emigrants to the OECD-6 countries accounting for a minimum of 75% of emigrants in OECD countries in 2000. This finding shows that the destination countries in our analysis receive proportionally more skilled migrants than the other OECD countries.

Maps 1a and 1b and 2a to 2d show that the “least reliable” countries are randomly distributed. In other words, they are not necessarily the smallest or poorest countries. The countries with the lowest proportion of migrants in the six main OECD receiving countries are in Latin America (although rates of skilled emigration are over 70% reliable for all the countries of South American except Brazil). The Democratic Republic of Congo, Angola and Namibia also have low levels of reliability, as have Morocco and Mauritania when total emigration rates are considered (the rate for skilled emigration from Morocco is 75.4% reliable).

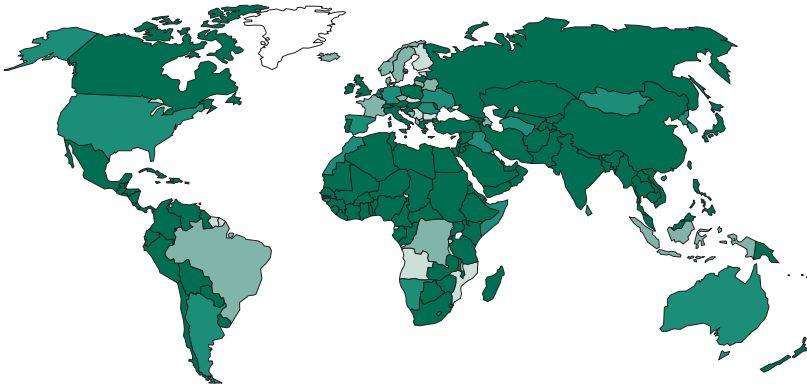
Last, new receiving countries have emerged in southern Europe since the mid-1990s, chiefly Italy, Spain and Portugal. The rapid growth in migrant flows to these countries could be expected to generate a large bias in our database. In fact, however, the bias is relatively small for two reasons. First, as we are working on the period 1975-2000, the bias should only affect the last year of the database. Second, since we work on migrant stocks and not migrant flows, the large increase in flows to the southern Europe countries in the period 1995-2000 has only a limited impact on stocks in 2000. The flow of new migrants is largely “diluted” in the stock. The new international migration flows are included in the database of Docquier and Marfouk (2005), who treat all the OECD countries as receiving countries. But although the large increase in these flows is taken into account, the rates for skilled and total emigration remain extremely stable over the period 1990-2000, confirming the “dilution” of the increased flows in the emigration stocks of the sending countries.

In the remainder of this study, we consider the countries of which at least 75% of emigrants to OECD countries reside in the six main receiving countries.

Map 1a. Proportion of emigrants aged 25 or over resident in one of six main OECD receiving countries in 2000 (as percentage of OECD total)



Map 1b. Proportion of highly skilled emigrants aged 25 or over resident in one of six main OECD receiving countries in 2000 (as percentage of OECD total)



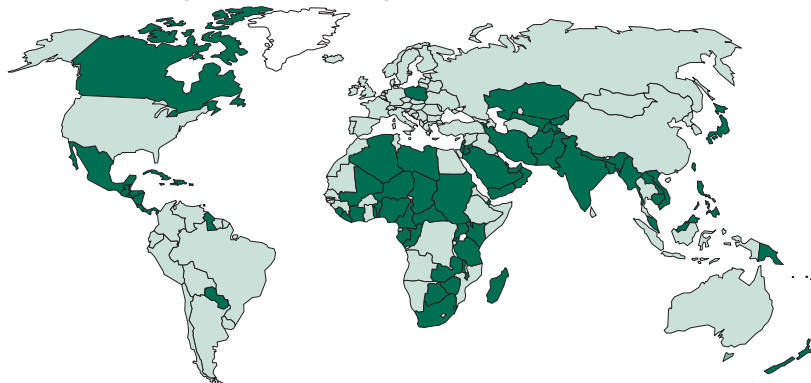
80-100% 70-80% 50-70% 0-50% n.d.

Interpretation: over 80% of highly skilled adults aged 25 or over who emigrated from Mexico to an OECD country reside in one of the six main OECD receiving countries in 2000 (Australia, Canada, United States, United Kingdom, Germany).

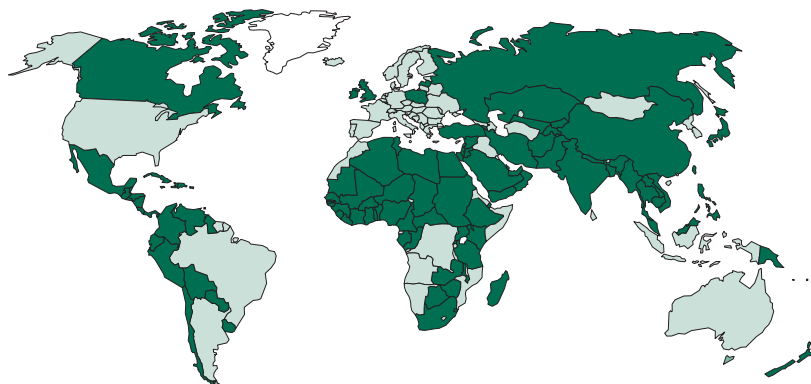
Source: Author's calculations based on census data from OECD-6 countries and Docquier and Marfouk (2004).

Map 2. Countries whose proportion of emigrants aged 25 or over resident in one of the six main OECD receiving countries exceeds 80% and 70% in 2000 (as percentage of OECD total)

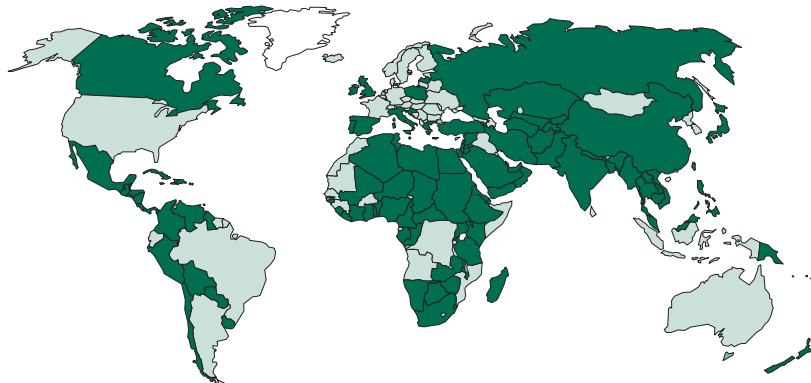
a. Proportion of migrants above 80% (in green)



b. Proportion of skilled migrants above 80% (in green)



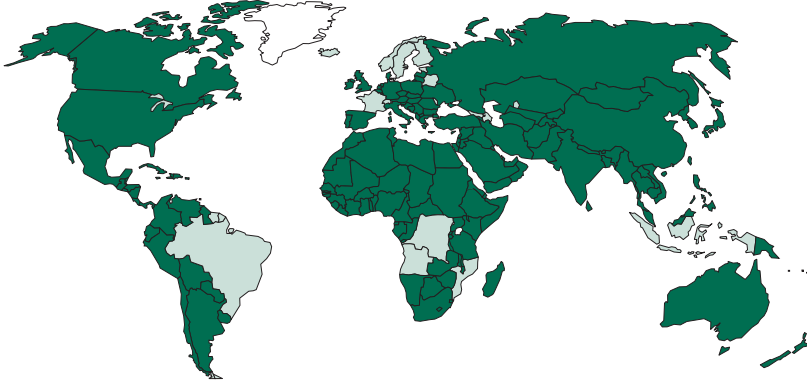
c. Proportion of migrants above 70% (in green)



Source: Author's calculations based on census data from OECD-6 countries and Docquier and Marfouk (2004).

Map 2 (cont'd). Countries whose proportion of emigrants aged 25 or over resident in one of the six main OECD receiving countries exceeds 80% and 70% in 2000 (as percentage of OECD total)

d. Proportion of skilled migrants above 70% (in green)



Source: Author's calculations based on census data from OECD-6 countries and Docquier and Marfouk (2004).

3. Estimating emigration rates by skill level in the period 1975-2000

When all the information on immigrant numbers by skill levels and country of origin in the main OECD receiving countries has been assembled, we aggregate these numbers to obtain emigration by country of origin. In order to work with emigration rates rather than stocks, the resident population of the country of origin is then introduced, distinguishing between the three skill levels described earlier. This approach enables us to make a comparison between the skilled emigrant and skilled resident populations, and hence to measure the impact of departures on the local labour market. We denote $N_{s,t}^j$ the number of individuals aged 25 or over, of skill level s , resident in country j , at time t . The total emigration rate is given by:

$$m_{s,t}^j = \frac{M_{s,t}^j}{N_{s,t}^j + M_{s,t}^j}$$

More specifically, $m_{s,t}^j$ supplies information about the size of the brain drain in sending country j . The data on skills of resident population aged 25 or over come from Barro and Lee (2000) for the developing countries, and from De la Fuente and Domenech (2002) for the OECD countries.

By introducing the populations observed in each period in the different sending countries, we can distinguish between resident and native populations. The native populations are all the individuals present in the country of origin before migration by a certain number of workers (pre-migration situation),

while the stock of residents corresponds to the population actually present in the sending country, once departure of the migrants is taken into account. The proportion of skilled workers among the residents, $n_{h,t}^j$, is given by:

$$n_{h,t}^j = \frac{N_{h,t}^j}{N_{h,t}^j + N_{m,t}^j + N_{l,t}^j}$$

II. Trends in human capital and international migration, 1975-2000

A database covering an extended period is of value chiefly for analysing long-term trends. Our database provides estimates of skill levels and emigration rates in the period 1975-2000 and can thus be used to analyse the brain drain during the last thirty years. The number of international migrants grew considerably between 1975 and 2000, more than doubling according to UN estimates. As noted earlier, unlike the UN, we consider only individuals aged 25 or over, but the overall change observed is the same. According to our data, the total number of adult international migrants in the six main OECD receiving countries rose from nearly 20 million in 1975 to over 44 million in 2000, a more than two-fold increase. The growth was even stronger for migration by highly skilled persons, whose number increased four-fold between 1975 and 2000 (nearly 4.3 million skilled international migrants in 1975 against nearly 17.3 million in 2000, equivalent to annual growth of 5.7%). However, it is noteworthy that the number of skilled workers in resident populations was also increasing over the same period in most countries of the world. Table 3.1 presents the change in the share of highly skilled workers in the total resident population by geographical sub-region.⁽⁹⁾ At the global level, this proportion rose from 5.6% in 1975 to 11.4% in 2000. However, the increase affected different countries in different ways.

The regions with the lowest proportion of skilled workers are generally in Africa (to a lesser extent in North African countries), but also include certain Asian countries and Pacific Islands.⁽¹⁰⁾ In these extremely poor regions, the proportion of skilled labour in the resident population was around 3% in 2000. The highest skill levels are observed in the most developed OECD countries, notably in North America, EU-15 and in Australia and New Zealand. In the other countries, the proportion of skilled labour in the resident population is between 5% and 8%. But are the countries with highest emigration rates of skilled workers those with the lowest share of skilled workers in the resident population?

(9) See Appendix for the composition by country of the geographical zones.

(10) The Pacific Islands comprise Melanesia, Micronesia and Polynesia.

Table 3.1. Percentage of highly skilled among resident population aged 25 or over, 1975-2000

Region	1975	1980	1985	1990	1995	2000
America	15.8	19.2	21.7	23.9	28.0	29.6
North America	26.6	32.3	36.2	39.7	47.8	51.3
Caribbean	2.1	3.4	4.9	6.6	8.4	9.3
Central America	3.1	5.1	6.9	8.8	10.0	11.1
South America	4.5	5.8	7.5	9.2	11.0	12.3
Europe	8.5	9.9	11.7	14.4	16.1	18.4
Eastern Europe	5.9	6.1	7.4	8.8	14.8	17.4
Rest of Europe	9.1	10.7	12.7	15.5	17.0	19.1
EU-15	9.1	10.8	12.7	15.5	17.4	19.5
EU-25	8.7	10.2	12.0	14.7	16.3	18.4
Africa	1.6	2.0	2.0	2.6	3.4	4.0
North Africa	1.9	3.1	3.1	4.8	6.1	7.5
Sub-Saharan Africa	1.5	1.6	1.7	1.8	2.5	2.8
Asia	2.6	3.0	3.7	4.7	5.5	6.3
East Asia	2.9	3.3	3.8	4.9	5.5	6.3
Central and southern Asia	2.2	2.3	3.3	3.7	4.5	5.0
Southeast Asia	2.3	3.1	3.9	5.3	6.6	7.9
West Asia	2.9	4.7	5.6	8.0	9.8	11.4
Oceania	20.3	23.2	24.8	25.5	27.1	27.8
Australia and New Zealand	23.5	27.0	28.9	29.8	31.8	32.7
Other Pacific countries	1.0	1.0	1.6	1.6	2.3	3.1

Source: Author's calculations based on Barro and Lee (2000) and De la Fuente and Domenech (2002).

Table 3.2. Emigration rates of highly skilled adults aged 25 or over to the six main OECD receiving countries, 1975-2000 (%)

Region of origin	1975	1980	1985	1990	1995	2000
America	2.0	2.0	2.5	2.6	2.5	3.0
North America	0.9	0.9	1.1	0.9	0.7	0.8
Caribbean	54.2	47.3	45.5	42.2	38.0	42.3
Central America	13.8	10.8	12.6	13.1	15.2	16.7
South America	3.6	3.5	3.8	3.7	3.4	4.2
Europe	8.6	8.4	7.9	7.1	4.3	5.3
Eastern Europe	7.1	8.6	8.5	8.9	8.9	9.8
Rest of Europe	8.8	8.3	7.7	6.9	5.6	6.4
EU-15	8.6	8.1	7.7	6.9	5.6	6.2
EU-25	8.8	7.4	6.1	4.8	4.0	3.4

Table 3.2 (cont'd). Emigration rates of highly skilled adults aged 25 or over to the six main OECD receiving countries, 1975-2000 (%)

Region of origin	1975	1980	1985	1990	1995	2000
Africa	7.3	7.6	9.5	8.8	9.0	9.3
North Africa	9.9	8.2	9.2	6.6	6.5	6.3
Sub-Saharan Africa	6.2	7.3	9.7	10.7	10.8	11.7
Asia	3.8	4.2	4.9	4.7	4.5	5.1
East Asia	2.2	2.4	3.1	3.1	3.1	3.4
Central and southern Asia	3.3	3.9	3.8	4.1	4.0	5.0
Southeast Asia	9.2	9.4	11.3	10.1	9.2	9.2
West Asia	12.6	9.5	9.3	7.0	5.9	6.2
Oceania	3.7	3.8	4.7	4.6	5.1	6.0
Australia and New Zealand	3.2	3.3	3.8	3.6	4.0	4.8
Other Pacific countries	45.6	47.7	50.8	52.1	48.7	43.0

Source: Author's calculations based on census data for the OECD-6 countries, Barro and Lee (2000) and De la Fuente and Domenech (2002).

Table 3.3. Emigration rate of low skilled adults aged 25 or over to the six main OECD receiving countries, 1975-2000 (%)

Region of origin	1975	1980	1985	1990	1995	2000
America	1.3	1.3	1.6	1.7	2.7	2.5
North America	1.1	1.2	1.2	1.0	1.9	1.0
Caribbean	5.1	5.6	6.0	5.7	7.9	5.7
Central America	3.3	3.5	4.6	5.6	8.9	8.7
South America	0.2	0.2	0.2	0.2	0.3	0.2
Europe	3.0	3.9	3.8	3.6	3.2	2.6
Eastern Europe	1.8	2.3	1.9	2.2	2.5	2.0
Rest of Europe	3.2	4.3	4.2	3.8	4.1	3.5
EU-15	3.2	4.9	4.1	3.8	3.9	3.4
EU-25	1.3	1.3	1.4	1.5	1.5	1.6
Africa	0.3	0.8	0.7	0.7	0.7	0.6
North Africa	0.5	2.4	2.2	2.0	1.9	1.8
Sub-Saharan Africa	0.2	0.2	0.3	0.3	0.3	0.2
Asia	0.2	0.2	0.3	0.2	0.3	0.3
East Asia	0.1	0.1	0.1	0.1	0.2	0.1
Central and southern Asia	0.1	0.1	0.2	0.2	0.2	0.1
Southeast Asia	0.2	0.2	0.3	0.4	0.5	0.4
West Asia	2.7	3.3	2.9	2.0	2.4	2.1
Oceania	1.4	1.2	1.7	1.6	2.1	1.3
Australia and New Zealand	1.4	1.3	1.9	1.8	2.3	1.3
Other Pacific countries	1.4	0.7	1.3	1.1	1.7	1.2

Source: Author's calculations based on census data for the OECD-6 countries, Barro and Lee (2000) and De la Fuente and Domenech (2002).

Table 3.4. Ratios between emigration rates of highly skilled and low-skilled adults aged 25 or over, 1975-2000 (relative risk)

Region of origin	1975	1980	1985	1990	1995	2000
America	1.6	1.5	1.6	1.5	0.9	1.2
North America	0.8	0.7	0.9	0.8	0.4	0.8
Caribbean	10.7	8.4	7.6	7.4	4.8	7.4
Central America	4.1	3.1	2.8	2.3	1.7	1.9
South America	20.5	20.6	19.5	21.0	10.6	19.2
Europe	2.9	2.1	2.1	2.0	1.3	2.0
Eastern Europe	4.0	3.8	4.5	4.0	3.5	5.0
Rest of Europe	2.7	1.9	1.9	1.8	1.4	1.9
EU-15	2.7	1.7	1.9	1.8	1.5	1.8
EU-25	6.9	5.6	4.4	3.2	2.7	2.1
Africa	27.7	10.0	13.0	13.2	13.2	15.9
North Africa	19.5	3.4	4.1	3.3	3.4	3.6
Sub-Saharan Africa	33.7	33.3	38.6	42.3	38.9	54.9
Asia	17.8	17.8	19.0	19.7	14.6	20.0
East Asia	30.4	36.2	36.2	31.2	19.4	32.5
Central and southern Asia	22.6	27.7	25.2	26.4	23.2	34.5
Southeast Asia	45.8	41.9	33.2	27.1	17.6	22.2
West Asia	4.7	2.9	3.3	3.6	2.5	2.9
Oceania	2.5	3.3	2.7	2.9	2.5	4.6
Australia and New Zealand	2.2	2.5	2.0	2.1	1.7	3.6
Other Pacific countries	32.3	68.5	37.7	46.9	29.4	35.4

Source: Author's calculations based on census data for the OECD-6 countries, Barro and Lee (2000) and De la Fuente and Domenech (2002).

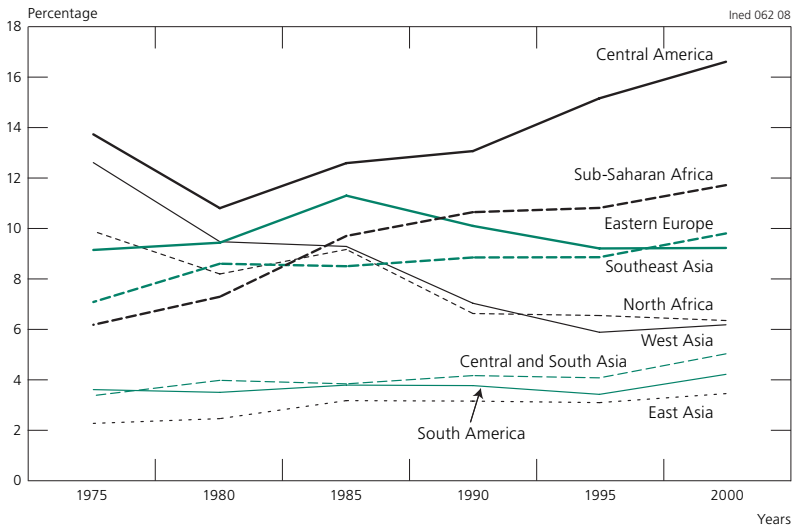
Tables 3.2, 3.3, and 3.4 indicate, respectively, the emigration rates for highly skilled and low-skilled adults, and the ratio between these rates (relative risk) for each world region, between 1975 and 2000. The countries where the emigration rate increased between 1975 and 2000 for both the highest and lowest skilled workers were in Central and South America, Eastern Europe, sub-Saharan Africa, and Asia (excepting West Asia). In the other world regions, the emigration rate for skilled individuals increased while for the low skilled it declined.

Emigration rates for highly skilled workers, which can be used to measure the scale of the brain drain, are on average relatively low in South America (4.2% in 2000) and East Asia (3.4%) but are close to 10% in Eastern Europe, 12% in sub-Saharan Africa, and 17% in Central America. Emigration rates for highly skilled workers are at their highest, albeit declining, in the Caribbean and the small Pacific Islands (42% and 43% respectively in 2000). Disregarding these two regions,⁽¹¹⁾ Figure 2 shows that the countries most affected by the brain

(11) The islands of the Caribbean and Pacific are not shown on the figure because they are too small for meaningful comparison with the other world regions. In addition, for many skilled immigrants from the Pacific the skills were probably obtained in the destination country.

drain are in Central America, sub-Saharan Africa and Eastern Europe, and that the phenomenon increased sharply after the 1980s. In other regions the skilled migration rate remained stable over time. This was the case in Southeast Asia (around 9.2%), Central and Southern Asia (between 3.3% and 5%), East Asia (2.2-3.4%) and South America (between 3.4% and 4.2%). Last, the brain drain declined in the other countries over the period 1975-2000, i.e. in North Africa (decline from 9.9% to 6.3%) and West Asia (from 12.6% to 6.2%).

Figure 2. Trends in emigration rates to OECD-6 of highly skilled adults aged 25 or over for the main world regions affected by brain drain

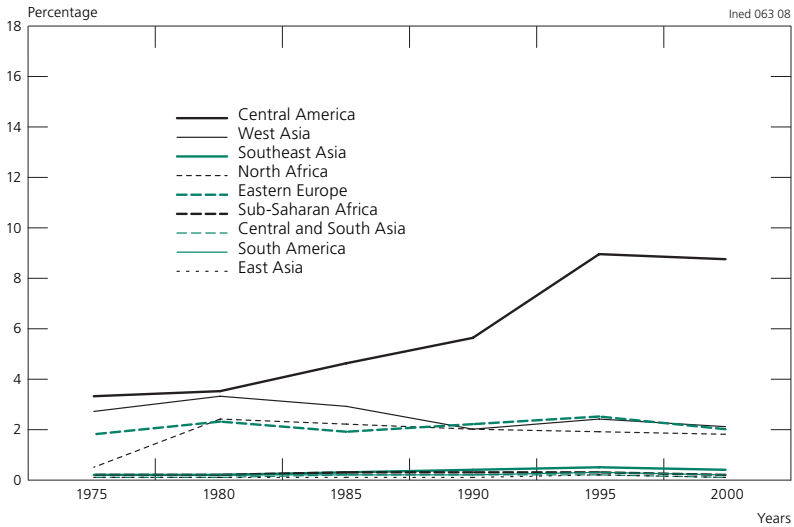


Source: Author's calculations based on census data for the OECD-6 countries, Barro and Lee (2000) and De la Fuente and Domenech (2002).

The emigration rates of low skilled adults remained fairly stable, except in the countries of Central America and North Africa (Figure 3). This stability is due to the sharp growth in the unskilled population among residents in the poorest regions of the world (note that the emigration rates are the ratios of emigrant numbers to resident populations for a given skill level). For the countries where an increase in the unskilled emigration rate was registered, the explanation may in part be the easing of the liquidity constraint: when this is less burdensome, larger numbers of poor people (low educational level) can leave their country of origin. In Central America and North Africa, average emigration rates for the unskilled rose from 3.3% to 8.7%⁽¹²⁾ and 0.5% to 1.8%, respectively. This rate held extremely stable in Eastern Europe, West Asia,

(12) This phenomenon is also explained by the sharp increase in migration to the United States by Mexicans, most of them unskilled.

Figure 3. Trends in emigration rates of low-skilled adults aged 25 or over to OECD-6 for the main world regions affected by brain drain



Source: Author's calculations based on census data for the OECD-6 countries, Barro and Lee (2000) and De la Fuente and Domenech (2002).

Southeast Asia, South America, sub-Saharan Africa, Central Asia, and East Asia.

The ratio between the skilled and unskilled emigration rates (Table 3.4) declined in many regions. The only regions where emigration by skilled adults increased relative to unskilled emigration were Eastern Europe, sub-Saharan Africa, East Asia, Central and South Asia, and the countries of Oceania (Australia and New Zealand). In the regions where the relative risk declined overall between 1975 and 2000, it nonetheless increased considerably between 1995 and 2000. The sharp acceleration in international migration by highly skilled individuals – and hence in the brain drain – since the mid-1990s can be attributed to rising levels of education in the poorest countries.

Last, Table 4 summarizes the main destination countries of migrants according to their region of origin in 2000. The majority of emigrants originating from Central and South America head for North America (and primarily the United States), whereas emigrants from Europe migrate to the group of six receiving countries considered, though no preferred destination stands out. Of migrants from North Africa, 74% went to France in 2000, whereas the United Kingdom was the preferred destination for those from sub-Saharan Africa (34%). The United States was also the main destination for individuals from Asia (53% of Asian migrants choose the US and the proportion even reaches 68% for migrants from East and Southeast Asia. Lastly, Australia is

Table 4. Resident population and migrants aged 25 and over in the OECD countries and the six main receiving countries, by region of origin, 2000

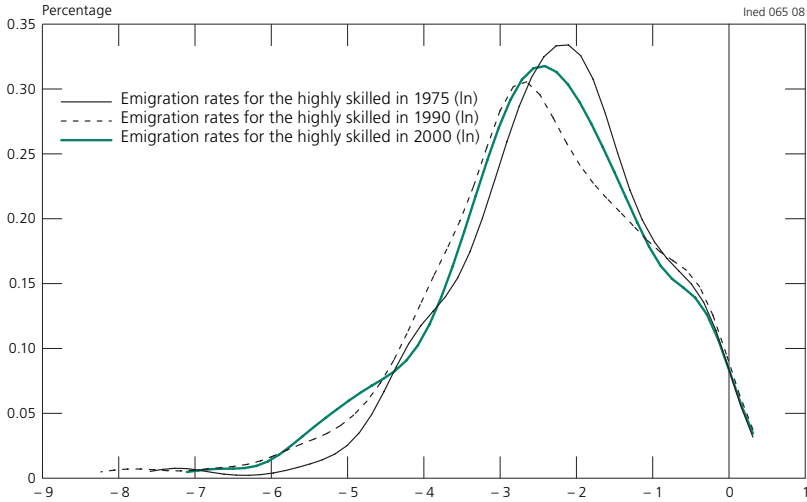
Region of origin	Resident population	Emigrant numbers in OECD countries	Emigrant numbers in OECD-6 countries	Distribution of emigrants present in OECD-6 countries (%)					
				United States	Canada	Australia	United Kingdom	France	Germany
America	453,776,796	15,517,891	13,981,493	89.6	4.9	0.9	3.0	0.6	1.0
North America	204,369,000	1,636,809	1,238,158	57.8	16.1	4.9	12.0	2.5	6.7
Caribbean	16,732,796	3,011,284	2,905,403	83.6	8.5	0.1	7.0	0.5	0.3
Central America	59,725,000	8,081,375	7,981,295	98.8	0.9	0.1	0.1	0.1	0.1
South America	172,950,000	2,788,422	1,856,637	81.0	8.9	2.8	3.1	2.0	2.2
Europe	499,479,352	21,094,702	13,759,125	30.7	15.0	13.9	7.7	13.0	19.6
Eastern Europe	201,629,000	4,635,945	2,505,862	47.3	15.2	4.9	3.0	6.0	23.5
Rest of Europe	297,850,352	16,458,758	11,253,263	27.0	15.0	15.9	8.8	14.5	18.7
EU-15	267,448,000	13,570,380	9,737,172	27.7	16.0	16.5	9.3	15.7	14.7
EU-25	316,342,000	16,235,347	11,446,492	29.2	16.1	15.5	9.2	14.4	15.6
Africa	298,221,989	4,496,950	3,201,429	20.0	7.0	4.5	19.2	44.6	4.7
North Africa	75,189,000	2,278,258	1,545,769	10.4	5.0	2.4	3.6	73.7	4.9
Sub-Saharan Africa	223,032,989	2,218,691	1,655,660	28.9	8.9	6.4	33.8	17.6	4.4
Asia	1,917,997,929	15,042,824	12,592,076	52.9	12.6	7.1	10.4	3.5	13.5
East Asia	910,164,929	4,171,944	3,274,391	68.3	17.7	6.6	4.2	1.4	1.8
Central and southern Asia	673,490,000	3,564,753	3,155,619	44.7	15.1	5.0	27.2	1.9	5.9
Southeast Asia	248,496,000	4,150,018	3,630,509	68.3	10.7	10.3	4.2	4.2	2.3
West Asia	85,847,000	3,156,108	2,531,557	21.1	5.3	5.7	6.5	7.2	54.2
Oceania	17,867,987	808,806	644,769	20.2	6.4	52.9	19.8	0.5	0.1
Australia and New Zealand	14,921,000	565,815	488,480	14.2	4.4	55.3	25.3	0.6	0.2
Other Pacific countries	2,946,987	242,991	156,289	38.9	12.8	45.6	2.5	0.2	0.0

Source: Author's calculations based on census data and Docquier and Marfouk (2004).

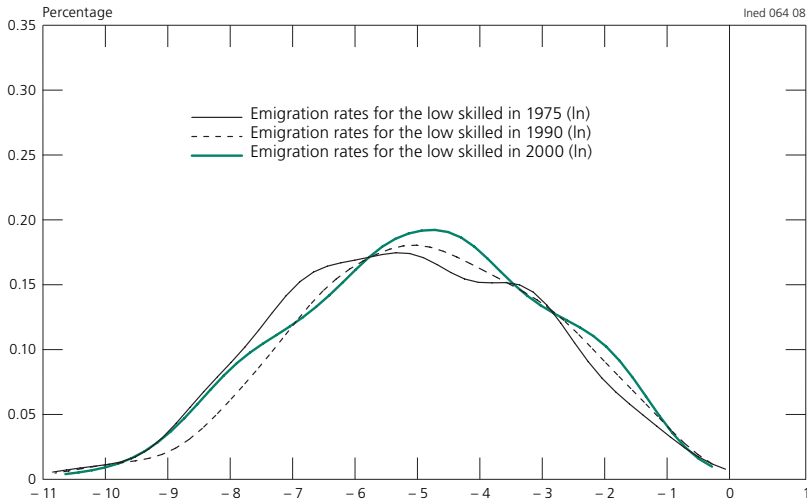
the main destination for migrants from Oceania, followed by the United States and the United Kingdom.

Figure 4. Density of emigration rates for adults aged 25 and over to OECD-6 countries, 1975, 1990, and 2000

a. Highly skilled



b. Low skilled



Source: Author's calculations based on census data for OECD-6, Barre and Lee (2000), and De la Fuente and Domenech (2002).

In the next section we focus on the trends in skilled international migration using a country-based approach.

III. Trends in skilled emigration rates by country

If we focus on rates of skilled emigration for all the countries in the world, Figure 4a is particularly instructive. It represents the density of skilled emigration rates for all the countries in the study (for which the rates are considered reliable), comparing these distributions in 1975, 1990 and 2000. Figure 4b depicts the density of unskilled emigration rates.⁽¹³⁾

Since 1975, there has been an increase in the number of countries where the emigration rate for the highly skilled – the “brain drain rate” – is low (below 1%) or comparatively high (between 1.5% and 30%), but a decrease in the number where it is over 30% (Figure 4a). At the same time, a fall is observed in the number of countries where the emigration rate of low-skilled adults is close to zero (below 0.2%), between 3% and 8%, or over 22% (Figure 4b). On the other hand, there was an increase in countries where the unskilled emigration rate is approximately between 0.2% and 3% or between 3% and 8%. This reflects the general rise in mobility levels throughout the world, made possible by ever lower migration costs and by the growth of migrant networks, since the existence of networks in destination countries tends to encourage migration by the low skilled (McKenzie and Rapoport, 2007).

Maps 3a and 3b represent country-level skilled emigration rates to OECD-6. The first point to note is that the countries most affected by brain drain are broadly the same in 2000 as in 1975. This applies specifically to the African countries (mainly sub-Saharan), the Caribbean, the Pacific Islands, and Central American countries. Africa and South America are shown in slightly lighter tones in Map 3b, suggesting a reduction in the brain drain in some countries despite an increase at regional level. This result reflects the widening disparities between countries in the African continent: since 1975, the situation for the worst-affected countries has deteriorated while for the others it has improved.

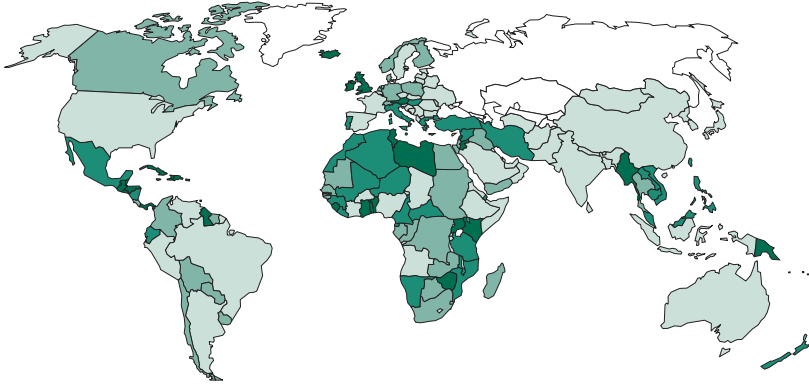
Maps 3a and 3b highlight a number of countries in particular. Emigration by the highest skilled workers affects some countries of the world much more severely than others. Table 5 presents the thirty countries (for which the data are considered “reliable”) with the highest skilled emigration rates to OECD-6 in 1975, 1990 and 2000.

Overall there is little change in the list of countries between the three years under consideration. The countries most often represented in the “top 30” are from the Caribbean and Latin America: Guyana, Barbados, Dominica, Grenada, Jamaica, Belize, Haiti, etc. Rates of skilled emigration to the six main OECD

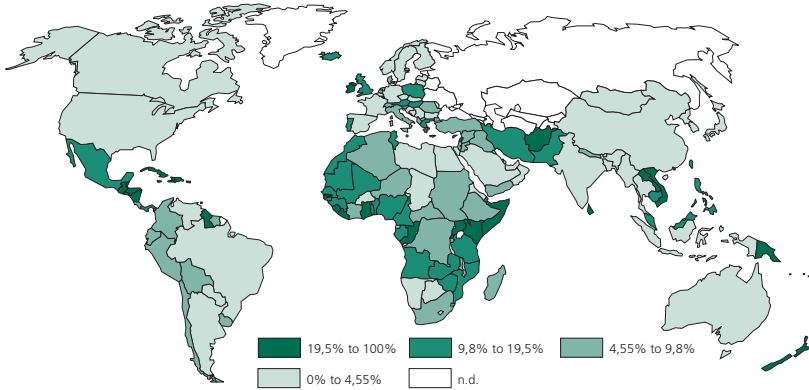
(13) These distributions are Gaussian.

Map 3. Emigration rates of highly skilled adults aged 25 and over in OECD-6

a. In 1975



b. In 2000



Source: Author's calculations based on census data for OECD-6, Barro and Lee (2000), and De la Fuente and Domenech (2002).

receiving countries have stood well above 50% for all of these countries since 1975. The list also contains African countries, such as Uganda and Ghana. The Pacific Islands – notably Fiji, Marshall Islands, and the Federated States of Micronesia – are also severely affected by the phenomenon. Although these tiny states are ranked at the bottom of the list, they experience a heavy loss of their most skilled workers, with emigration rates ranging between 35% and 60% depending on period. Despite the high stability exhibited by the ranking, it is noteworthy that Cuba, Panama and Hong Kong are no longer among the most affected countries in 2000. States such as Laos and Sierra Leone, on the other hand, which were not among the thirty most affected countries in 1975, have since joined the list, with skilled emigration rates of over 30% in 2000.

Table 5. Countries most affected by brain drain to OECD-6 in 1975, 1990, and 2000

Emigration rates of highly skilled adults aged 25 or over (%)					
Top 30 in 1975	%	Top 30 in 1990	%	Top 30 in 2000	%
Guyana	87.7	Guyana	91.0	Guyana	88.9
Barbados	85.7	Jamaica	85.1	Grenada	85.1
Dominica	85.4	St-Vincent-&-Grenadines	80.7	Jamaica	85.0
Saint Kitts and Nevis	85.3	Grenada	80.6	St-Vincent-&-Grenadines	84.5
Grenada	82.6	Haiti	78.4	Haiti	83.5
Jamaica	81.6	Saint Kitts and Nevis	77.8	Trinidad and Tobago	79.2
Trinidad and Tobago	81.1	Trinidad and Tobago	77.8	Saint Kitts and Nevis	78.5
Haiti	80.3	Gambia	76.7	Saint Lucia	71.0
Malta	79.9	Tonga	74.7	Tonga	69.6
St-Vincent-&-Grenadines	72.9	Malta	68.4	Antigua and Barbuda	66.7
Antigua and Barbuda	72.4	Dominica	68.3	Belize	65.4
Belize	71.7	Saint Lucia	67.6	Dominica	63.9
Bahamas	64.0	Barbados	67.2	Barbados	63.4
Nauru	62.3	Belize	66.9	Bahamas	61.1
Seychelles	59.8	Antigua and Barbuda	64.8	Gambia	58.8
Mauritius	57.9	Mauritius	63.7	Fiji	57.3
Saint Lucia	53.8	Fiji	60.5	Malta	56.8
Uganda	51.6	Seychelles	58.7	Seychelles	55.1
Tonga	46.0	Bahamas	57.2	Mauritius	55.0
Cuba	45.4	Palau	57.0	Sierra Leone	51.8
Ireland	43.5	Uganda	43.2	Ghana	45.1
Lebanon	41.6	Lebanon	42.4	Liberia	44.2
Jordan	39.5	Kenya	42.0	Marshall Islands	39.3
Micronesia	38.5	Nauru	41.4	Kenya	37.8
Fiji	38.4	Papua New Guinea	37.6	Micronesia	37.6
Togo	36.4	Ghana	34.9	Laos	37.2
Panama	35.8	Ireland	34.8	Lebanon	37.1
Cyprus	33.8	Micronesia	34.5	Uganda	34.7
Hong Kong	33.6	Sierra Leone	33.6	Eritrea	31.9
Kenya	33.4	Hong Kong	32.3	Nauru	31.3

Source: Author's calculations based on census data for OECD-6 countries, Barro and Lee (2000), and De la Fuente and Domenech (2002).

Table 6 presents the group of outliers. These are the countries for which the rate of skilled emigration to OECD-6 changed most – in either direction – between 1975 and 2000. The skilled emigration rate for Afghanistan was multiplied more than tenfold between 1975 and 2000, but very large increases were also registered for Liberia, Rwanda, Somalia, Laos, Sri Lanka, the Marshall Islands, and the Republic of Congo. These countries share a common feature: all are characterized by political instability or a climate of insecurity – both “push” factors that encourage the departure of skilled workers.⁽¹⁴⁾ The countries

(14) The importance of these “push” factors is highlighted by studies on the determinants of brain drain (for example, Docquier and Rapoport, 2005).

Table 6. Countries with largest variation in emigration rates for skilled adults aged 25 or over to OECD-6 between 1975 and 2000

Largest rate increases			Largest rate decreases		
	Rate (%)			Rate (%)	
	1975	2000		1975	2000
Afghanistan	1.8	21.3	Trinidad and Tobago	81.1	79.2
Rwanda	3.8	21.5	Malta	79.9	56.8
Somalia	5.7	27.0	Uganda	51.6	34.7
Liberia	11.8	44.2	Cuba	45.4	27.6
Laos	10.4	37.2	Nauru	62.3	31.3
Sri Lanka	7.9	28.0	Togo	36.4	18.2
Marshall Islands	11.5	39.3	Zimbabwe	24.3	11.7
Republic of Congo	7.0	20.9	Panama	35.8	15.5
Nicaragua	13.1	28.8	Tunisia	27.4	11.2
Gambia	28.9	58.8	Central African Rep.	17.4	6.9
Sierra Leone	27.9	51.8	Benin	30.2	11.0
Ghana	27.0	45.1	Niger	16.7	5.7
Tonga	46.0	69.6	Papua New Guinea	100.0	27.6
Fiji	38.4	57.3	Namibia	14.2	2.7
Saint Lucia	53.8	71.0	Burma	21.2	3.8
St-Vincent-&-Grenadines	72.9	84.5	Jordan	39.5	6.8
Jamaica	81.6	85.0	Lesotho	20.9	3.5
Haiti	80.3	83.5	Libya	20.3	2.2
Grenada	82.6	85.1	Maldives	10.1	0.9
Guyana	87.7	88.9	Bhutan	20.0	0.5

Source: Author's calculations based on census data for OECD-6 countries, Barro and Lee (2000), and De la Fuente and Domenech (2002)

where the rate of brain drain has fallen sharply since 1975 include Bhutan, the Maldives, Libya, Jordan, Lesotho, etc.

In the following section, we will examine the brain drain in the most affected countries, by population size.

IV. The scale of the brain drain: analysis by population size

When analysing the countries most affected by the departure of skilled workers, respective population sizes must be taken into account. For instance, a direct comparison between China (with nearly 1.3 billion inhabitants in 2000) and Grenada (with approximately 100,000 inhabitants at the same date) would be quite meaningless. Table 7 lists in decreasing order the 25 countries most affected by brain drain for three categories of country distinguished by number of inhabitants: under 2.5 million, between 2.5 and 20 million, and over 20 million in 2000. The skilled emigration rates to OECD-6 given in the

table are those for 1975 and 2000. Here again, only the countries for which the estimated skilled emigration rate is 75% reliable are used in the analysis.

The large countries with the highest levels of skilled emigration in 1975 are Uganda, Kenya and Burma, all of which have a rate exceeding 20% (and for Uganda, even over 50%). Lower down the list are the United Kingdom, Iran, certain countries of North Africa (Morocco and Algeria) and Asia (Vietnam and Malaysia). All these countries have a brain drain rate of between 10% and 20%. How does this list change between 1975 and 2000? We observe a high degree of stability: the majority of countries most affected in 1975 are still present in 2000, including those mentioned above. Noteworthy, however, is the arrival in the list of Ethiopia, Nepal, Nigeria, and Pakistan. This confirms an exacerbation of the brain drain problem affecting mainly the African continent.

Among countries with between 2.5 and 20 million inhabitants, the worst affected in 2000 are Jamaica, Haiti, Cuba, Lebanon, and Hong Kong. Compared with the large countries, the change observed in their case between 1975 and 2000 is more striking. Between 1975 and 2000, 10 countries joined the list, with emigration rates for skilled workers of over 20%: Sri Lanka, Somalia, Croatia, Laos, Liberia, Nicaragua, New Zealand, Dominican Republic, Democratic Republic of Congo, and Rwanda. On the other hand, three European countries – Austria, Greece, and Portugal – came off the list, resulting in a relative improvement in the situation of the richest countries compared with the developing countries.

The last section of Table 7 presents the 25 countries with below 2.5 million inhabitants and the highest skilled emigration rates. It contains mainly Caribbean and Pacific islands. The list of the worst affected countries in 1975 is practically identical in 2000 (though we note the arrival of the Marshall Islands and the Comoros).

All in all, these results reveal a negative correlation between the population size of countries and their rate of skilled emigration. Whereas the average value of the skilled emigration rates is close to 13% for the 25 large countries most affected in 1975 and 2000, it is around 34% for those of intermediate size and 55% for the small countries. International migration and the loss of skilled workers thus seem to weigh more heavily on the smaller countries, whose resident populations contain too few skilled workers to make up for the shortages created by those who leave.⁽¹⁵⁾ Compensating for the loss of skilled workers is obviously easier for large countries like China or India, since it is hard to see how their reserves of skilled labour could ever be exhausted.

(15) For these countries, however, the high rates are probably explained by the absence of educational infrastructures, which causes individuals to migrate for study purposes.

Table 7. Countries with highest brain drain in 1975 and 2000, by population size in 2000
(emigration rates of highly skilled adults aged 25 or over to OECD-6 countries expressed as a percentage)

Population over 20 million in inhabitants			Population between 2.5 and 20 million inhabitants			Population below 2.5 million inhabitants					
Country	1975	Country	2000	Country	1975	Country	2000	Country	1975	Country	2000
Uganda	51.6	Kenya	37.8	Papua New Guinea	100.0	Jamaica	85.0	Guyana	87.7	Guyana	88.9
Kenya	33.4	Uganda	34.7	Jamaica	81.6	Haiti	83.5	Barbados	85.7	Grenada	85.1
Burma	21.2	Vietnam	26.3	Haiti	80.3	Sierra Leone	51.8	Dominica	85.4	St Vincent	84.5
United Kingdom	19.9	Afghanistan	21.3	Cuba	45.4	Ghana	45.1	Saint Kitts and Nevis	85.3	Trinidad and Tobago	79.2
Iran	17.8	Mexico	15.2	Ireland	43.5	Liberia	44.2	Grenada	82.6	Saint Kitts and Nevis	78.5
Algeria	17.1	United Kingdom	14.3	Lebanon	41.6	Laos	37.2	Trinidad and Tobago	81.1	St Lucia	71.0
Vietnam	16.1	Morocco	13.3	Jordan	39.5	Lebanon	37.1	Malta	79.9	Tonga	69.6
Morocco	15.1	Philippines	13.2	Togo	36.4	Eritrea	31.9	St Vincent	72.9	Antigua and Barbuda	66.7
Italy	12.7	Iran	13.0	Panama	35.8	El Salvador	30.4	Antigua and Barbuda	72.4	Belize	65.4
Malaysia	12.4	Poland	12.7	Hong Kong	33.6	Nicaragua	28.8	Antigua and Barbuda	71.7	Dominica	63.9
Turkey	11.5	Tanzania	12.0	Honduras	31.1	Ireland	28.6	Bahamas	64.0	Barbados	63.4
Mexico	11.5	Pakistan	12.0	Benin	30.2	Hong Kong	28.4	Nauru	62.3	Bahamas	61.1
Tanzania	10.4	Malaysia	10.3	Sierra Leone	27.9	Sri Lanka	28.0	Seychelles	59.8	Gambia	58.8
Philippines	9.9	Nigeria	10.1	Tunisia	27.4	Papua New Guinea	27.6	Mauritius	57.9	Fiji	57.3
Iraq	9.0	Colombia	9.3	Ghana	27.0	Cuba	27.6	St Lucia	53.8	Malta	56.8
Poland	8.5	Romania	9.2	Zimbabwe	24.3	Somalia	27.0	Tonga	46.0	Seychelles	55.1
Colombia	7.9	Ethiopia	9.2	El Salvador	22.0	Honduras	23.9	Micronesia	38.5	Mauritius	55.0
Egypt	7.5	Algeria	8.6	Libya	20.3	Guatemala	23.5	Fiji	38.4	Marshall Islands	39.3
South Africa	6.0	Iraq	8.6	Guatemala	20.1	Croatia	21.6	Cyprus	33.8	Monroesia	37.6
Thailand	5.8	Italy	8.4	Austria	19.7	Rwanda	21.5	Gambia	28.9	Nauru	31.3
Sudan	5.7	South Africa	6.8	Cameroon	19.1	Dominican Rep.	20.9	Kiribati	23.6	Cyprus	30.8
Canada	4.7	Sudan	6.4	Malawi	18.0	Rep. of Congo	20.9	Macao	21.3	Macedonia	27.3
Peru	4.5	Turkey	5.2	Portugal	17.7	New Zealand	20.4	Lesotho	20.9	Palau	25.6
Romania	4.4	Peru	4.9	Central African Rep.	17.4	Malawi	18.3	Palau	20.1	Comoros	21.1
Korea	4.4	Nepal	4.7	Greece	17.2	Togo	18.2	Bhutan	20.0	Kiribati	21.0
Mean	13.2		13.1		35.1		33.3		55.8		55.8

Source: Author's calculations based on census data for OECD-6 countries, Barro and Lee (2000) and De la Fuente and Domenech (2002).

Source: Author's calculations based on census data for OECD-6 countries, Barro and Lee (2000) and De la Fuente and Domenech (2002).

Conclusion

This article has presented a new database on international migration to the six main OECD receiving countries since 1975. It shows that skill levels at age 25 and over have risen across the world since 1975, with the proportion of skilled individuals in resident populations climbing from 5.6% to 11.4%. But this increase is very unevenly distributed, and skill levels in the poorest countries have remained particularly low. Concerning trends in skilled emigration rates between 1975 and 2000, our results show an increase in skilled emigration relative to unskilled emigration in many world regions (and in all regions during the decade 1990-2000). The brain drain is nonetheless taking place against a background of rising skill levels and increasing international mobility driven by globalization. Although the trend – upward or downward – was sizeable in thirty or so countries, emigration rates for a large number of countries show relative stability. However, this should not mask the fact that strong push factors are still operative in many countries, particularly those with small populations.

While past trends in international migration are helpful for understanding the difficulties currently facing sending countries, future trends will be decisive on several counts. First of all, demographic changes over the next fifty years will radically reshape the labour markets in origin and destination countries. Strong pressures will be exerted upon the economically active populations in the northern countries where the ageing process is most advanced, and where the temptation will be strong to recruit skilled labour from abroad. If increasingly selective policies are implemented in these countries, the impact on the countries of the South could be largely cushioned by their rapid population growth. But will they be able to meet the challenge of funding education for their native populations in response to demand from the North? In addition, the receiving countries will need to introduce effective integration policies for foreigners if they wish to avoid the problems now facing Europe, where governments are struggling to assimilate the second and third generation immigrants who still feel out of place in our societies.



APPENDIX

Definition of geographical zones

America: *North America*, (Canada, United States), *Caribbean*, (Antigua and Barbuda, Bahamas, Barbados, Cuba, Dominica, Dominican Republic, Grenada, Haiti, Jamaica, Saint Kitts and Nevis, Saint Lucia, St-Vincent-&-Grenadines, Trinidad and Tobago), *Central America*, (Belize, Costa Rica, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama), *South America*, (Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Guyana, Paraguay, Peru, Suriname, Uruguay, Venezuela).

Europe: *Eastern Europe*, (Bulgaria, Czech Republic, Hungary, Poland, Romania), *Rest of Europe*, (Denmark, Estonia, Finland, Iceland, Ireland, Latvia, Lithuania, Norway, Sweden, United Kingdom, Albania, Andorra, Bosnia-Herzegovina, Croatia, Greece, Vatican, Italy, Macedonia, Malta, Portugal, San Marina, Serbia-and-Montenegro, Slovenia, Spain, Austria, Belgium, France, Germany, Liechtenstein, Luxembourg, Monaco, Netherlands, Switzerland), *EU-15*, (Denmark, Finland, Ireland, Sweden, United Kingdom, Greece, Italy, Portugal, Spain, Austria, Belgium, France, Germany, Luxembourg, Netherlands), *EU-25*, (EU-15 + Estonia), Latvia, Lithuania, Malta, Slovenia, Czech Republic, Hungary, Poland, Slovakia, Cyprus).

Africa: *North Africa*, (Algeria, Egypt, Libya, Morocco, Sudan, Tunisia), *Sub-Saharan Africa*, (Burundi, Comoros, Djibouti, Eritrea, Ethiopia, Kenya, Madagascar, Malawi, Mauritius, Mozambique, Rwanda, Seychelles, Somalia, Tanzania, Uganda, Zambia, Zimbabwe, Angola, Cameroon, Central African Republic, Chad, Democratic Republic of Congo, Republic of Congo, Equatorial Guinea, Gabon, São Tomé-&-Príncipe, Botswana, Lesotho, Namibia, South Africa, Swaziland, Benin, Burkina Faso, Cape Verde, Côte d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania, Niger, Nigeria, Senegal, Sierra Leone, Togo).

Asia: *East Asia*, (China, Hong Kong, Macao, Japan, Korea, Mongolia, Taiwan), *Central and southern Asia*, (Afghanistan, Bangladesh, Bhutan, India, Iran, Kazakhstan, Kyrgyzstan, Maldives, Nepal, Pakistan, Sri Lanka, Tajikistan, Turkmenistan, Uzbekistan), *Southeast Asia*, (Brunei, Burma, Cambodia, Timor, Indonesia, Laos, Malaysia, Philippines, Singapore, Thailand, Vietnam), *West Asia*, (Armenia, Azerbaijan, Bahrain, Georgia, Iraq, Israel, Jordan, Kuwait, Lebanon, Palestine, Oman, Qatar, Saudi Arabia, Syria, Turkey, United Arab Emirates, Yemen).

Oceania: *Australia and New Zealand*, (Australia, New Zealand), *Other Pacific islands*, (Fiji, Papua New Guinea, Solomon Islands, Vanuatu, Kiribati, Marshall Islands, Federated states of Micronesia, Nauru, Palau, Samoa, Tonga, Tuvalu).

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CÉCILY DEFOORT • LONG-TERM TRENDS IN INTERNATIONAL MIGRATION: AN ANALYSIS OF THE SIX MAIN RECEIVING COUNTRIES

This article proposes an estimation of international migration to the six main receiving countries of the OECD between 1975 and 2000. The analysis reveals a large increase in international migration during this period, together with a substantial change in migrant skills. In parallel with this phenomenon we note a significant rise in skill levels at world level. An analysis by country shows a high degree of stability in the list of the countries most affected by the brain drain. While overall change in the migration of skills is limited, the list of countries with the largest losses of highly skilled workers in 1975 is largely unchanged twenty-five years later.

CÉCILY DEFOORT • TENDANCES DE LONG TERME DES MIGRATIONS INTERNATIONALES : ANALYSE À PARTIR DES SIX PRINCIPAUX PAYS RECEVEURS

Cet article propose une estimation de la migration internationale en direction des 6 principaux pays receveurs de l'OCDE de 1975 à 2000. L'analyse révèle une hausse importante des migrations internationales au cours de la période, accompagnée d'une évolution importante de la qualification des migrants. Parallèlement à ce phénomène, nous notons une hausse significative du niveau de qualification au niveau mondial. Une analyse par pays montre une forte stabilité du classement des pays les plus touchés par la fuite des cerveaux. Si l'évolution globale des migrations qualifiées s'avère limitée, les pays étant les plus touchés par le départ de leurs travailleurs les plus qualifiés en 1975 demeurent majoritairement les mêmes vingt-cinq ans plus tard.

CÉCILY DEFOORT • TENDENCIAS DE LARGO PLAZO DE LAS MIGRACIONES INTERNACIONALES : ANÁLISIS A PARTIR DE LOS SEIS PRINCIPALES PAÍSES RECEPTORES

Este artículo propone una estimación de la migración internacional dirigida a los 6 principales países receptores de la OCDE de 1975 a 2000. El análisis revela un alza importante de las migraciones internacionales durante el período, junto con una evolución importante de la cualificación de los migrantes. Paralelamente a este fenómeno, notamos un alza significativa del nivel de cualificación a nivel mundial. Un análisis por país muestra una fuerte estabilidad de la clasificación de los países más afectados por la fuga de cerebros. Si la evolución global de las migraciones cualificadas resulta limitada, los países más afectados por la salida de sus trabajadores más cualificados en 1975 siguen siendo mayoritariamente los mismos veinticinco años más tarde.